

Bayesian Belief Network

- graphical model that represents probabilistic relationships among set of variables. BBNs are used for reasoning under uncertainty, where relationships betⁿ variables are modeled using probabilistic dependencies.

Here's how it works:-

1) Nodes:-

The var in BBNs are represented as nodes in graph. Each node represents a random variable, which can be discrete or continuous & has associated probability distⁿ.

2) Edges:-

- they connectⁿ betⁿ nodes.
- encode probabilistic dependencies betⁿ variables.

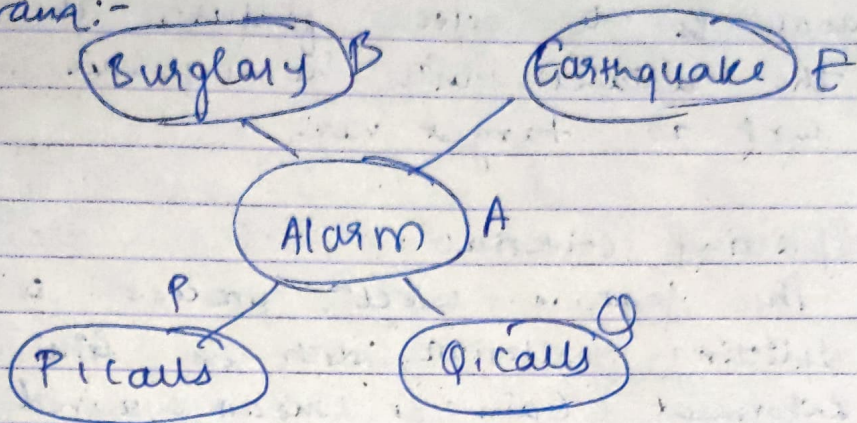
3) Conditional Probability Table (CPTs):-

Each node in network has a conditional probability table associated with it, which specifies probabilities of node's possible values given the values of its parent nodes.

4) Inference:-

Once the Bayesian Belief Network is constructed, it can be used for probabilistic inference. Given observed evidence for certain variables, network can compute probabilities of other var.

Diagram:-



* Decision Tree Induction

- ML technique used for classification & regression tasks.
- It builds a tree-like structure where each internal node represents a decision based on value of a feature, each branch represents outcome of that decision, and each leaf node represent final decision or prediction.

* Concepts:-

1) Root Node:-

- represents entire dataset.

2) Feature selection:-

- At each internal node, a decision is made based on value of a chosen feature. The

goal is to select feature the best splits the dataset into homogenous subsets wrt to target var.

3) Splitting criterion

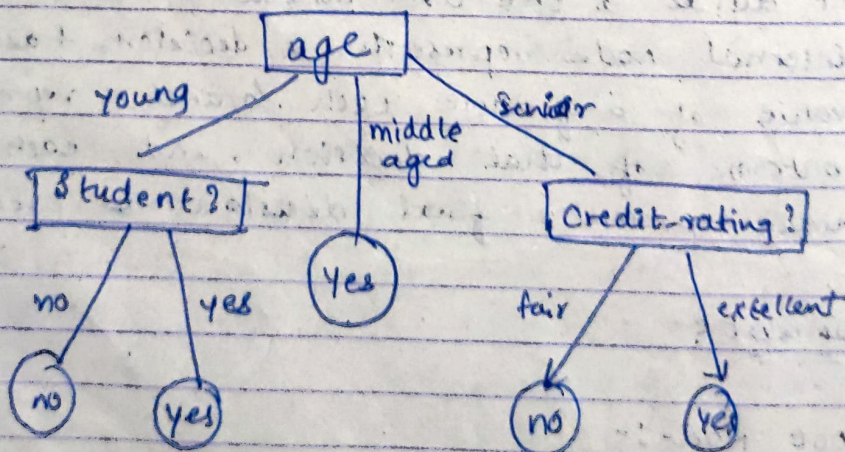
The feature selectⁿ process is guided by splitting criterion, such as Gini impurity, or Informative Gain or mean squared error.

The criterion measures impurity or uncertainty of data set before & after the split, A goal is to minimize it.

4) Leaf Node :-

Leaf Node represents final decision or prediction. Once splitting criterion is met, no further splits are made.

Eg with Diagram



The foll. decision tree is for concept ^{to} buy-computer that indicates whether a customer or a company is likely to buy a comp. or not. each internal node represents a test on an attribute. each leaf node represents class.

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* Compare & contrast strength & weaknesses of Bayesian Network & Decision tree.

Strength

Bayesian Network	Decision Tree
1/ BBNs provide transparent, interpretable models that are easy to understand & explain.	1/ Decision tree offer human readable representation of decision-making processes.
2/ Can handle missing data effectively by incorporating probabilistic reasoning.	2/ can handle missing values by making decision based on available data at each node.
3/ can capture non-linear relationships bet ⁿ variables through complex probabilistic dependencies.	3/ can model non-linear relationships directly without requiring feature transformations.

Weaknesses

1/ can become computationally expensive.	1/ can become complex & computationally intensive.
2/ may struggle to handle irrelevant features effectively.	2/ can automatically identify & ignore irrelevant feature.
3/ prone to overfitting, especially when network structure has insufficient data to estimate parameters accurately.	3/ are susceptible to overfitting particularly with deep trees that capture noise in data. Regularization techniques are required to solve this issue.

Goals of Decision-Tree Induction CIFH

Goal - is to create predictive model that can accurately classify or predict outcomes based on input features.

Here are some obj:-

- 1) Classification & Prediction:-
 - main purpose of decision tree is to build a model that can classify instances into predefined categories or predict numerical values based on input features.
- 2) Interpretability:-
 - designed to be interpretable, meaning that the model's decision-making process is easy to understand or explain.
- 3) Feature Importance:-
 - aims to identify the most imp. features or variable for predicting the target variable.
- 4) Handling Missing Data:-
 - aims to handle missing data effectively by making decisions based on available data at each node.

* Bayesian Belief Network real-world scenario & its applications:-

⇒ SCENARIO:- Diagnosing a medical condition:-

• Variables:-

1. Symptoms:- Categorical (Fever, cough, Headache, fatigue)
2. Test Results:- Categorical (Positive, Neg)
3. Medical History:- Categorical (Yes, NO)
4. Family History:- cate (Yes, NO)
5. Diagnosis:- categorical (Diag A, Diag B, No disease)

Assumptions:-

- Have certain symptoms increases likelihood of a +ve test result.
- A +ve test result increases likelihood of having the disease.
- Previous medical history & family hist. may influence likelihood of have disease.

Application:-

- Symptoms:- fever, cough
- Test result:- +ve
- Medical hist:- No
- Fam hist:- Yes

We can use BBNs to predict likelihood of patient have Disease A, Disease B, or No Disease.

Impact of attribute selection measures on constructⁿ of decision tree.

Inf. Gain:-

Impact:-

measures effectiveness of an attribute in reducing uncertainty about target variables. Attributes with higher inf. gain are preferred for splitting nodes as they provide more useful inf. for classification, effective when dealing with attributes that have multiple possible values or categories.

Gini Index:-

Impact:-

measures impurity of a dataset, with lower values indicating greater purity. Attributes with lower Gini Index values are preferred for splitting nodes, as they result in more homogeneous child nodes. Effective in situations where goal is to minimize impurity within each node of decision tree.

Impact on D.T.:-

- Both I.G & G.I guide DT. process by determining attribute to split on at each node.
- help identify most informative attributes for partitioning the dataset, leading to creation of decision tree that effectively discriminate between diff. classes.
- I.G may be preferred when goal is to maximize class separation, while G.I may be preferred when goal is to minimize impurity within nodes.

Distances

Q. 6

ID	TV 1	TV 2
1	20	A
2	30	B
3	20	A
4	10	C

✓ $\frac{p-m}{p} \Rightarrow$ Nominal Attribute Proximity.

here $p = 2$.

$$p(1, 2) = \frac{2-0}{2} = 1$$

$$p(1, 3) = \frac{2-2}{2} = 0$$

$$p(1, 4) = \frac{2-0}{2} = 1$$

$$p(2, 1) = 1$$

	1	2	3	4
1	0	1	0	1
2	1	0	1	1
3	0	1	0	1
4	1	1	1	0

$$p(2, 3) = 1$$

$$p(2, 4) = 1$$

$$p(4, 3) = 1$$

✓ Euclidean distance. $(x_1, y_1) = (2, 3.5)$

$$d(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(1.5 - 2)^2 + (1.6 - 3.5)^2}$$

$$= \sqrt{(0.25) + (3.61)}$$

$$= \sqrt{3.86}$$

$$= 1.96$$

(1.4, 1.8)

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Q4

Manhattan

$$(x_1, y_1) = (1.4, 1.6)$$

$$\sqrt{x_2^2 + y_2^2}$$
$$|x_2 - x_1| + |y_2 - y_1|$$

$$|1.5 - 1.4| + |1.7 - 1.6|$$

$$|0.1| + |0.1|$$

$$\Rightarrow |0.2| = 0.2$$

Minkowski

$$\left[\sum_{i=1}^n |x_i - y_i|^p \right]^{1/p}$$

$$\left[24, 2, 43, 9 \right] \quad \left[20, 0, 36, 8 \right]$$

$p=3$

$$\Rightarrow \left[(24-20)^3 + (2-0)^3 + (43-36)^3 + (9-8)^3 \right]^{1/3}$$

$$\left[(4)^3 + (2)^3 + (7)^3 + (1)^3 \right]^{1/3}$$

$$\left[64 + 8 + 343 + 1 \right]^{1/3}$$

$$\left[416 \right]^{1/3}$$

$$\Rightarrow \boxed{7.42}$$